Global Psychological & Health Analytics

Empirical Trends in Suicide Mortality, Behavioural Risk Factors and Life Outcomes (2015 – 2025)

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Tooling: Supabase PostgreSQL, Google Looker Studio, Google Colab

Executive Summary

This project explores the relationship between adult tobacco use and suicide mortality rates across different countries using World Bank World Development Indicators (WDI) data from 2015 to 2025. The main aim was to investigate whether countries with higher levels of adult tobacco use also tend to have higher suicide mortality rates, and whether tobacco prevalence could potentially be used as a broad behavioural indicator of psychological distress or suicide risk. The data was cleaned and transformed using Supabase PostgreSQL and then analysed visually in Google Looker Studio using a dual-axis model. Overall, the analysis did not show a clear direct or linear relationship between adult tobacco prevalence and national suicide mortality rates.

Analytical Justification & Problem Statement

Substance use is often discussed in clinical and psychological literature as a possible way that people cope with psychological distress, trauma, or socio-economic stress. Since these factors can also contribute to suicide risk, I wanted to explore whether a commonly available behavioural measure—adult tobacco prevalence—could provide any meaningful indication of suicide mortality at a national level.

The analysis was guided by three main questions:

Can national tobacco prevalence be used as an indirect indicator of country-level suicide risk?

Is suicide mortality directly proportional to adult tobacco prevalence?

Do countries with high levels of tobacco use consistently experience higher suicide mortality, or are there other contextual factors that may protect populations from this outcome?

The purpose was therefore not to suggest that tobacco use causes suicide, but rather to test whether the two variables show a meaningful relationship at a macro level.

Dataset Description

The data was sourced from the World Bank World Development Indicators (WDI) database and covered the period from 2015 to 2025 across 265 countries and regional entities.

The indicators used were:

Suicide mortality rate (per 100,000 population)

Prevalence of current tobacco use among adults (%)

Life expectancy at birth, total (years)

Mortality rate, adult, male (per 1,000 male adults)

Mortality rate, adult, female (per 1,000 female adults)

These variables allowed the analysis to look at tobacco prevalence alongside suicide mortality, while also providing broader mortality and population health indicators for additional context.

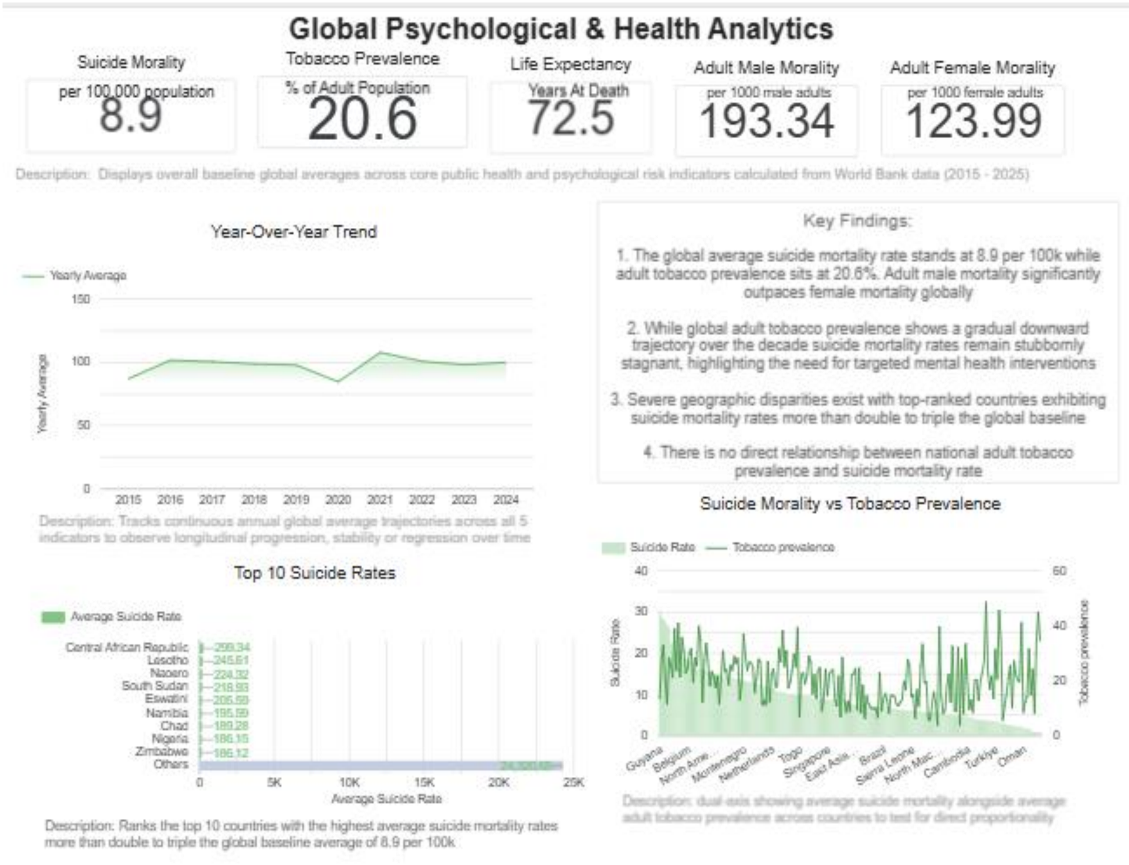
Data Engineering & Analytics Pipeline

The raw WDI data was first imported into Supabase and stored in a staging table

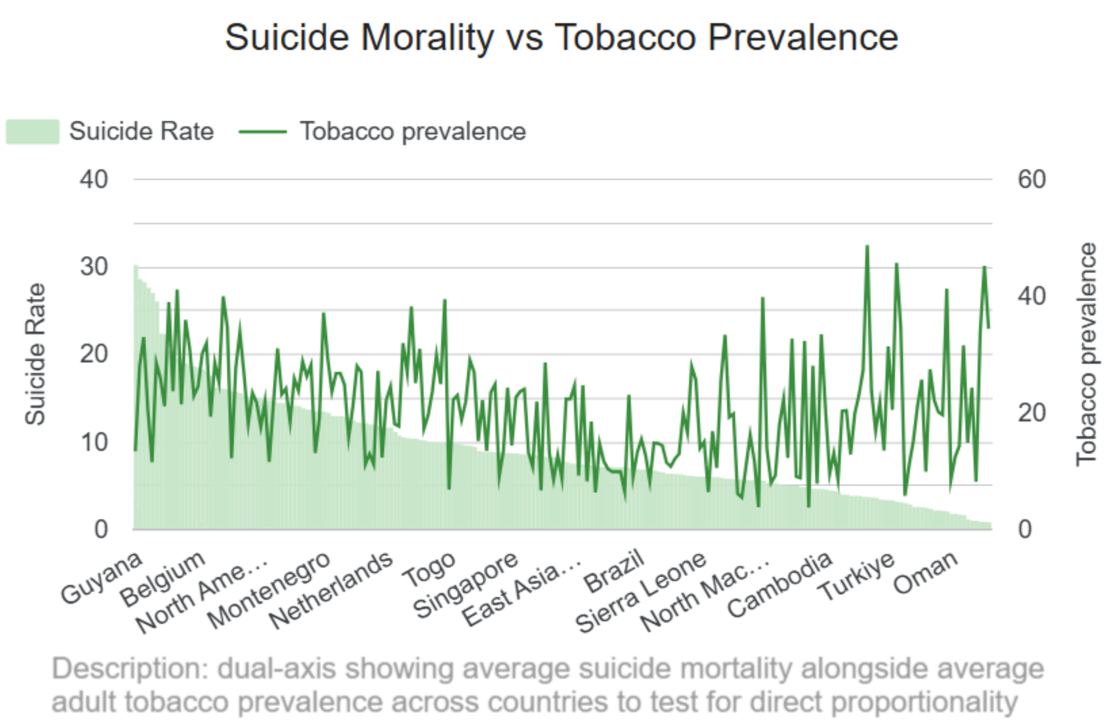
The original dataset stored each year as a separate column, therefore I used PostgreSQL to restructure the data into a more usable format. A view called clean_wdi_health was created using UNION ALL to convert the annual columns from 2015–2025 into a single year_val field. This made it easier to analyse the data across time.

The cleaned data was then connected to Google Looker Studio. An inner join was used to create the Tobacco_vs_Suicide_Blend, matching countries based on country_name. This allowed the analysis to focus on countries where concurrent records were available for both tobacco prevalence and suicide mortality.

For the main visual analysis, I created a dual-axis Combo Chart. Suicide mortality was represented using descending columns on the primary axis, while tobacco prevalence was displayed as a line on the secondary axis. This made it possible to visually assess whether the two measures appeared to move together.



Global Psychological & Health Analytics
Suicide Morality
per 100,000 population
8.9
Tobacco Prevalence
% of Adult Population
20.6
Life Expectancy
Years At Death
72.5
Adult Male Morality
per 1000 male adults
193.34
Adult Female Morality
per 1000 female adults
123.99
Description: Displays overall baseline global averages across core public health and psychological risk indicators calculated from World Bank data (2015 - 2025)
Year-Over-Year Trend
Yearly Average
150
100
50
0
Yearly Average
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
Description: Tracks continuous annual global average trajectories across all 5 indicators to observe longitudinal progression, stability or regression over time
Key Findings:
1. The global average suicide mortality rate stands at 8.9 per 100k while adult tobacco prevalence sits at 20.6%. Adult male mortality significantly outpaces female mortality globally
2. While global adult tobacco prevalence shows a gradual downward trajectory over the decade suicide mortality rates remain stubbornly stagnant, highlighting the need for targeted mental health interventions
3. Severe geographic disparities exist with top-ranked countries exhibiting suicide mortality rates more than double to triple the global baseline
4. There is no direct relationship between national adult tobacco prevalence and suicide mortality rate
Suicide Morality vs Tobacco Prevalence
Suicide Rate
Tobacco prevalence
Top 10 Suicide Rates
Average Suicide Rate
Central African Republic
Lesotho
Naoero
South Sudan
Eswatini
Namibia
Chad
Nigeria
Zimbabwe
Others
0
5K
10K
15K
20K
25K
Average Suicide Rate
Description: Ranks the top 10 countries with the highest average suicide mortality rates more than double to triple the global baseline average of 8.9 per 100k



Suicide Morality vs Tobacco Prevalence
Suicide Rate
Tobacco prevalence
40
30
20
10
0
Suicide Rate
60
40
20
0
Tobacco prevalence
Guyana
Belgium
North Ame…
Montenegro
Netherlands
Togo
Singapore
East Asia…
Brazil
Sierra Leone
North Mac…
Cambodia
Turkiye
Oman
Description: dual-axis showing average suicide mortality alongside average adult tobacco prevalence across countries to test for direct proportionality

Key Findings & Empirical Insights

1. No clear proportional relationship

The main finding was that there was no obvious linear or directly proportional relationship between adult tobacco prevalence and suicide mortality.

When countries were ordered according to suicide mortality, tobacco prevalence did not follow the same pattern. Instead, the tobacco prevalence line moved quite erratically, with both high- and low-tobacco countries appearing across different levels of suicide mortality.

This suggests that simply having a high national prevalence of tobacco use does not necessarily correspond with a high national suicide mortality rate.

High tobacco prevalence does not always correspond with high suicide mortality

A few countries illustrated this particularly well:

Lebanon: Adult tobacco prevalence was approximately 44.94%, while the suicide mortality rate was only 0.74 per 100,000.

Timor-Leste: Tobacco prevalence was approximately 48.52%, while suicide mortality was 3.56 per 100,000.

Bulgaria: Tobacco prevalence was approximately 39.28%, alongside a moderate suicide mortality rate of 9.76 per 100,000.

These examples demonstrate why tobacco prevalence cannot be treated as a straightforward proxy for suicide risk. If the relationship were directly proportional, countries with some of the highest tobacco prevalence would consistently be expected to have some of the highest suicide mortality rates. The data did not support this.

2. Limitations of using a single behavioural proxy

One of the broader insights from this analysis is that behavioural indicators need to be interpreted within their social and cultural context.

While tobacco use may be associated with coping, stress, or other individual-level factors, national tobacco prevalence is also influenced by factors such as cultural norms, affordability, availability, legislation, marketing, and social acceptance.

This means that a high national smoking rate cannot automatically be interpreted as evidence of high levels of psychological distress or suicidality.

3. Gender differences in mortality

The additional mortality indicators also showed a consistent pattern of higher adult male mortality compared with adult female mortality across most regions.

This provides an important gender-related context when thinking about suicide and broader mortality outcomes. It also highlights the importance of considering gender-specific patterns in mental health research, particularly when examining help-seeking behaviour, social expectations, and barriers to accessing psychological support.

Recommendations & Policy Implications

Move away from using broad behavioural measures as suicide warning signals

Public health authorities should be cautious about treating national tobacco prevalence, or similar broad behavioural measures, as indirect warning signals for suicide risk.

The findings suggest that these indicators are too heavily influenced by social, cultural, and structural factors to function as standalone measures of national suicidality.

Focus more on contextual and protective factors

Future research should look beyond individual behavioural indicators and examine the factors that may either increase or protect against suicide risk.

These could include:

Community and social connectedness

Social safety nets

Cultural and community resilience

Access to mental healthcare and psychiatric services

Economic conditions

Social isolation

Help-seeking behaviours

Stigma surrounding mental health

This may provide a much more useful picture of why countries with similar levels of tobacco use can experience very different suicide mortality outcomes.

Use gender-sensitive approaches to mental health support

The consistent differences in male and female mortality also reinforce the importance of gender-sensitive mental health interventions.

Mental health programmes should consider the ways in which social expectations, stigma, and traditional ideas around masculinity can affect men's willingness to seek psychological support. Outreach strategies that specifically engage men may therefore be important when addressing suicide prevention and broader mental health outcomes.

Conclusion

This project used PostgreSQL data transformation and Google Looker Studio visual analytics to investigate whether adult tobacco prevalence could be used as a broad indicator of national suicide mortality.

The findings did not support the hypothesis of a direct or linear relationship between the two variables. Countries with high tobacco prevalence did not consistently have high suicide mortality rates, demonstrating that tobacco prevalence alone is not a reliable proxy for national suicide risk.

More broadly, the analysis reinforces the idea that suicide is a complex, multifactorial outcome. Behavioural measures such as tobacco use may provide useful information when considered alongside other variables, but they should not be interpreted in isolation.

Understanding suicide mortality therefore requires a more contextual approach—one that considers psychological, social, cultural, economic, and protective factors together rather than relying on a single behavioural indicator.